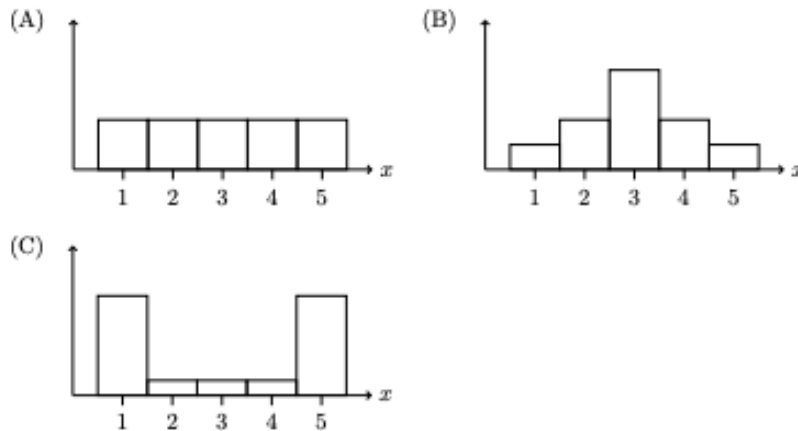


Quiz #2

1. The graphs below show the probability density functions of three random variables.



- (a) Order them by mean from smallest to largest.

They are all the same.

- (b) Order them by variance from smallest to largest.

BAC

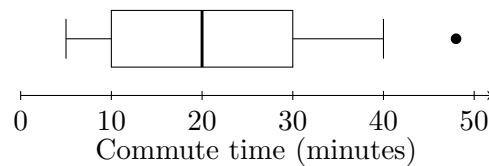
- (c) Order them by standard deviation from smallest to largest.

BAC

2. True or False: If $\sigma^2(X) = 0$, where σ^2 is the variance of X , then X is constant.

True

3. The boxplot below summarizes the one-way commute times (in minutes) for a group of students.



- (a) Approximately what is the median commute time?

About 20 minutes.

- (b) Approximately what interval contains the middle 50% of commute times?

About 10 to 30 minutes. These are the first and third quartiles, so the box contains the middle 50% of the observations.

- (c) What does the point near 48 minutes indicate? What does the boxplot suggest about the shape of the distribution?

The point near 48 minutes is a potential high outlier. The longer upper whisker together with the high outlier suggests that the distribution is right-skewed.

4. In words, explain what this code does. How many plots are produced, and what is shown in each plot?

```
variables = ["height", "weight"]

for variable in variables:
    plt.hist(df[variable], bins=10)
    plt.title(variable)
    plt.show()
```

The loop runs once for "height" and once for "weight". It produces two histograms: one showing the distribution of height and one showing the distribution of weight, each using 10 bins. The title of each plot is the name of the variable being plotted.